Contact
Industrial Cables
Phone: 845-469-2141
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ULTREX® VN Unshielded Tray Cable - 3 or 4 Conductor With Ground

ULTREX® VN Unshielded Tray Cable 3 Conductors With Ground

Part Number: VN With Ground

UL Type TC-ER, THHN / THWN-2 Insulation, 600 V, rated 90°C dry and wet – RoHS Compliant - Exposed Run (ER) Rated

Description

Applications

Nexans 600 V ULTREX® VN Tray Cables are listed as type TC-ER under UL 1277 Electrical Power and Control Cables. These cables may be installed in wet or dry locations; in cable trays, raceways and open air; and are suitable for exposure to weather, direct burial Class I, Div. 2 (also Zone 2) and Class II, Div. 2 hazardous locations, as well as Class III, Div. 1 & 2 hazardous locations per NEC. Cables with three or more conductors are UL listed for exposed runs (ER) when installed in accordance with NEC Article 336.10(7).

Construction

Conductor:
Bare, annealed copper conforming to ASTM B3 and Class B compressed stranded copper in accordance to ASTM B8, from 14 AWG to 500 kcmil.

Insulation:
Flame-retardant PVC/Nylon type THHN/THWN-2 per UL 83.

Assembly:
Cabled in concentric layers with interstices filled with suitable fillers, as required. Ground wires are sized as required by UL 1277 (refer to applicable product tables for the standard sizes provided). Sizes 14 AWG to 6 AWG have an insulated green ground wire. Size 4 AWG and larger have a bare ground wire. Where necessary, a binder tape of synthetic material assembles the core in a tight circular configuration.

Jacket:
UL listed sunlight and moisture resistant, sequentially marked, black, flame retardant polyvinyl chloride (PVC) material meeting the requirements of UL 1277. A Nylon ripcord is included for ease of jacket removal. Jacket surface is printed with required UL / NEC code information and sequential footage markings.

NEXANS-C ULTREX-VN 600V TC-ER 3/C-12 AWG THHN/THWN-2 SUN RES DIR BUR (UL) ROHS (mm/dd/yyyy)

Conductor Identification:
14 AWG to 10 AWG: Color coded per Method 1-E2 per ICEA S-73-532

8 AWG and larger: Black with number coding per Method 4 of ICEA S-73-532

See details in Conductor Identification Charts.

Standards

National ICEA S-73-532;ICEA S-95-658;UL 1277;UL 83

All drawings, designs, specifications, plans and particulars of weights, size and dimensions contained in the technical or commercial documentation of Nexans is indicative only and shall not be binding on Nexans or be treated as constituting a representation on the part of Nexans.

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Characteristics

Construction characteristics

Number of conductors	3

Unshielded ULTREX® VN Tray Cable - UL Type TC-ER, 600 V, 90°C dry and wet rated - RoHS Compliant

Type THHN/THWN-2: 3 Conductor with Ground

Part Number	Conductor Size (AWG/ kcmil)	Ground Size (AWG)	Insulation Thickness PVC/Nylon		Jacket Thickness		Nominal Diameter over Jacket		Approximate Net Cable Weight		Ampacity
			mils	mm	mils	mm	inches	mm	lb/kft	kg/km	amps
664995	14	14	15/4	0.38/0.10	45	1.14	0.350	8.89	90	134	25(1)
685578	12	12	15/4	0.38/0.10	45	1.14	0.368	9.35	126	188	30(1)
665000	10	10	20/4	0.51/0.10	45	1.14	0.45	11.43	202	301	40(1)
680747	8	10	30/5	0.76/0.13	60	1.52	0.610	15.49	302	449	55
680749	6	8	30/5	0.76/0.13	60	1.52	0.725	17.37	422	628	75
658948*	4	8	40/6	1.02/0.15	80	2.03	0.857	21.77	647	963	95
658955*	2	6	40/6	1.02/0.15	80	2.03	0.986	25.30	963	1435	130
680752	1	6	50/7	1.27/0.18	80	2.03	1.108	28.14	1156	1720	150
658963*	1/0	6	50/7	1.27/0.18	80	2.03	1.207	30.66	1451	2159	170
665018	2/0	6	50/7	1.27/0.18	80	2.03	1.297	32.94	1738	2586	195
216341	3/0	4	50/7	1.27/0.18	80	2.03	1.410	35.81	2122	3158	225
665026	4/0	4	50/7	1.27/0.18	80	2.03	1.530	38.86	2576	3833	260
665034	250	4	60/8	1.52/0.20	110	2.79	1.739	44.17	3128	4655	290
665042	350	3	60/8	1.52/0.20	110	2.79	1.961	49.81	4347	6469	350
665059	500	2	60/8	1.52/0.20	110	2.79	2.234	56.74	5819	8659	430

* DOW construction cables.

Bend Radius: 5 x overall diameter installed (up to 2.0") / 6 x overall diameter installed (over 2.0")
8 x overall diameter during installation pull-in.

Notes:
• Dimensions and weights shown are nominal values, subject to standard industry tolerances.
• Cables with different conductor counts and bare or insulated grounds are also available.
• Cables are UL listed for exposed runs (ER) when installed in accordance with NEC Article 336.10(7).
• Ampacities are in accordance with NEC Table 310.15(B)(16) for conductors in a raceway or direct buried at 30°C ambient temperature and 90°C rated conductors.
• For correction factors for different ambient temperatures and ampacities at different conductor temperature ratings see NEC Table 310.15(B)(16).
(1) NEC Article 240.4(D) requires that overcurrent protection not exceed 15 amperes for 14 AWG copper conductors, 20 amperes for 12 AWG, and 30 amperes for 10 AWG copper conductors after any correction factors for ambient temperature and number of conductors have been applied. Exceptions to this may be covered in NEC Article 240.4(E) through (G).

*Dow Construction means:
• Control conductors are color coded E-2
• Power conductors are black with numbers - M4
• Fillers where applicable.

Selling information

Options

The following constructions can be provided on special orders:

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* Dow Construction
* DuPont Construction
* Composite Cable Construction
* Different conductor identification methods

* Use in Hazardous locations:
Please note that no investigation of these cables has been performed regarding the transmission of gases or vapors through the core. When these cables are used in hazardous locations they should be sealed properly as required by the NEC.

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